BattleOps: Week 3

Theme: Kubernetes Security, Platform Hardening & Real-World Incident Prevention

Objective: Apply Kubernetes security concepts in real production-like situations, debug misconfigurations, enforce policies, and secure workloads end-to-end.

THURSDAY: Secrets Hardening, KMS, SealedSecrets

ADVANCED CONCEPTS

- Why Kubernetes Secrets are not real encryption
- Encrypting secrets at-rest using KMS
- Why GitOps requires encrypted secrets
- Using SealedSecrets / SOPs in production

KEY OUTCOMES

- You can configure and verify encryption-at-rest
- You can rotate secrets safely
- You can commit encrypted secrets to Git securely

WHAT TO PRACTICE

Lab 1: Enable KMS Encryption

Understand & apply the encryption config.

Lab 2: Install & Use SealedSecrets

Encrypt → commit → apply → decrypt in cluster.

Lab 3: Rotate Secrets

Demonstrate zero-downtime rotation.

WHERE TO STUDY

- Secrets encryption docs
- Bitnami SealedSecrets GitHub

INFRATHRONE BATTLEOPS NOTES

- Secrets must never exist unencrypted outside the cluster.
- Rotation is more important than creation.
